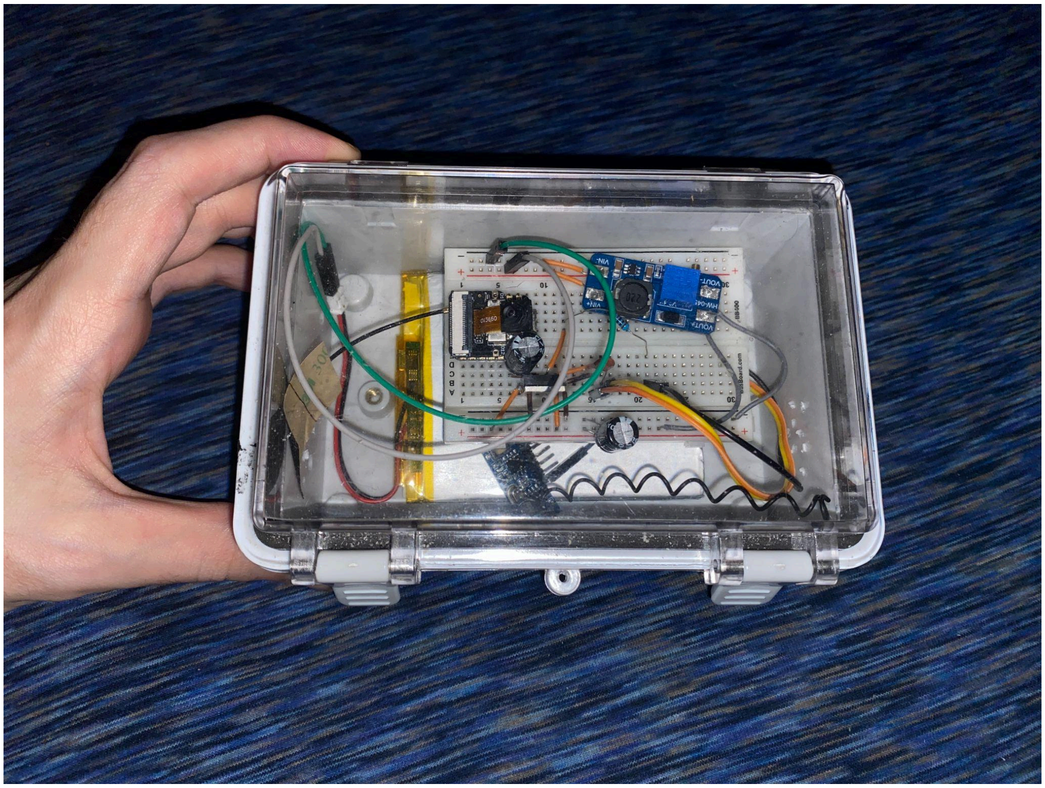
CAMRON

Extremely Efficient Remote Activated Wireless Camera With Solar Charging Capability

Main code with instructions
https://github.com/tddev123/Camron/tree/main

Youtube Video
https://www.youtube.com/watch?v=1h0qovzgmT0



PRIMARY COMPONENTS

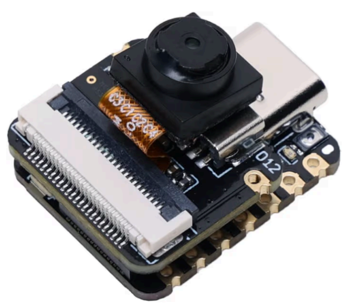
ESP32S3

seeed studio
Visit the Store 4.5 (720)

Seeed Studio XIAO ESP32 S3 Sense - 2.4GHz Wi-Fi, BLE 5.0, OV2640 Camera Sensor, Digital Microphone, 8MB PSRAM, 8MB Flash, Battery Charge Supported, Rich Interface, IoT, Embedded ML ...

Amazon's Choice

500+ bought in past month



3.7V 3000mAh

Qimoo
Visit the Store 4.3 (338)

407090 3.7V Lipo Battery 3000mAh 407090 Rechargeable Lithium Polymer Battery Pack with JST PH2.0mm Connector for Electronic Device

Waterproof Enclosure

Visit the Laisomeke Store 4.5 (60)

150x100x70mm ABS Junction Box with Hinged Clear Grey Cover Plastic Latch, IP67 Waterproof Enclosure with Mounting Plate Wall Bracket for Electrical Project 5.9x3.9x2.8 inch

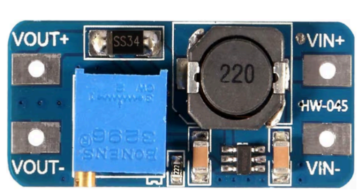
Overall Pick

MT3608

Visit the DORHEA Store 3.8 (481)

MT3608 DC-DC Step Up Boost Power Converter - 2A Module, Adjustable Step Up Voltage Regulator Board - 2-24V to 5V-28V Output Voltage (Pack of 10)

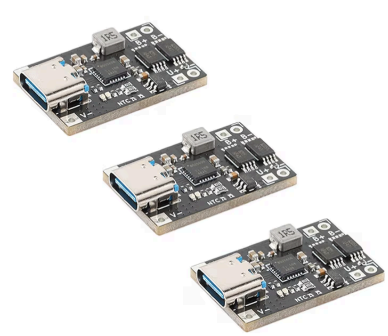
300+ bought in past month



BQ25606

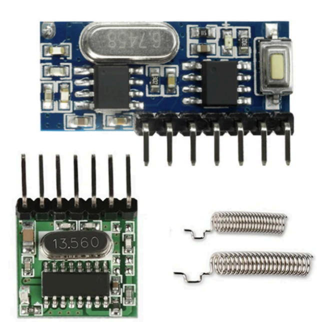
EC Buying
Visit the Store 3.4 (8)

3Pcs BQ25606 3A Lithium Battery Charging Module Type C 4.5-12V 3.7V Charging Protection Integrated Power Module Board



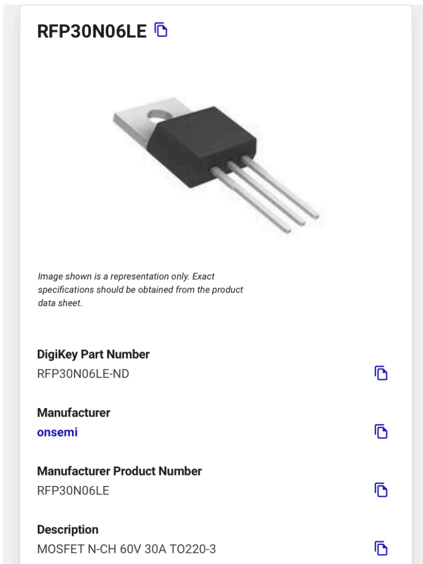
RX480E and TX118SA-4

Home › QIACHIP RX480E TX118SA-4 433.92Mhz Wireless Wide Voltage Coding Transmitter Module And Decoding Receiver Module 4CH RF EV1527 2262 Encoding Learning Module Diy Kit

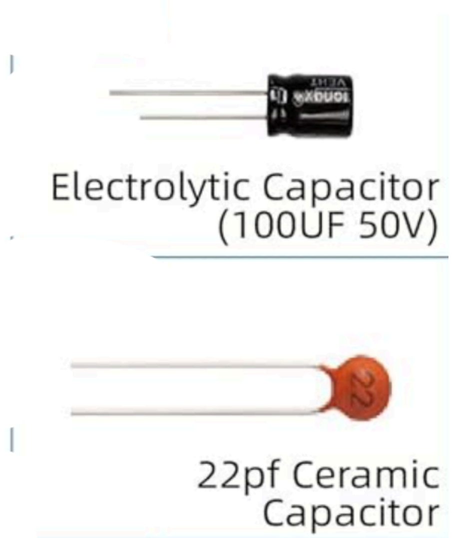


SECONDARY COMPONENTS

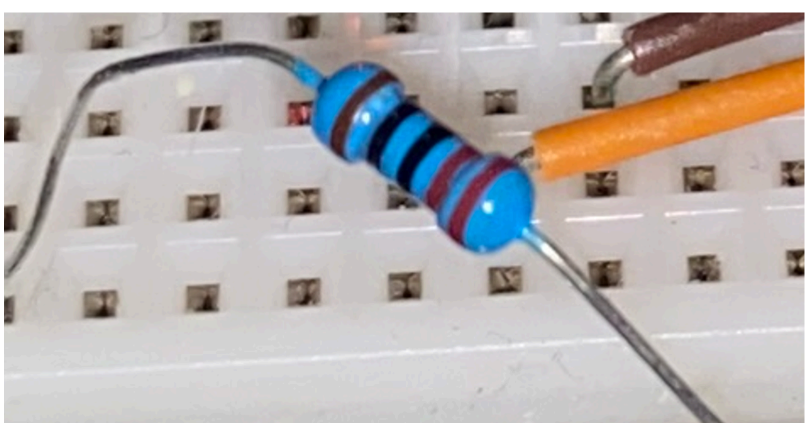
RFP30N06LE



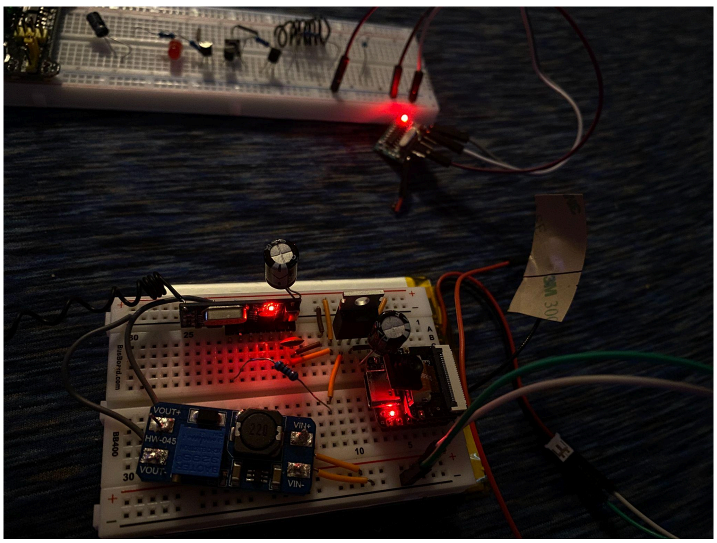
2 x 100 uF
1 x 22 pF

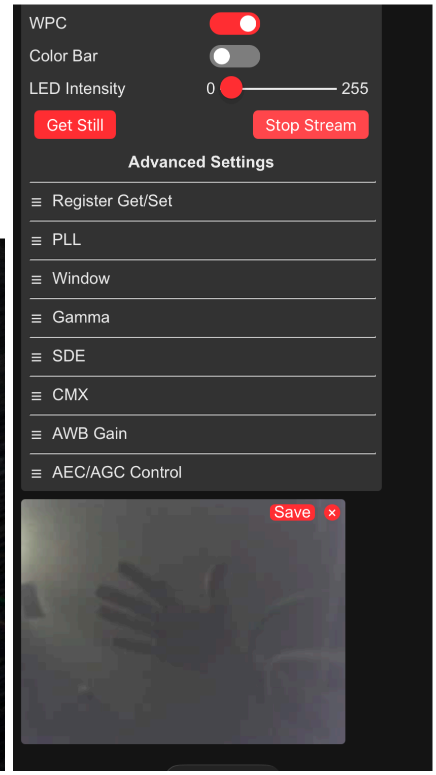


1 x 200 ohm



EVIDENCE OF FUNCTION



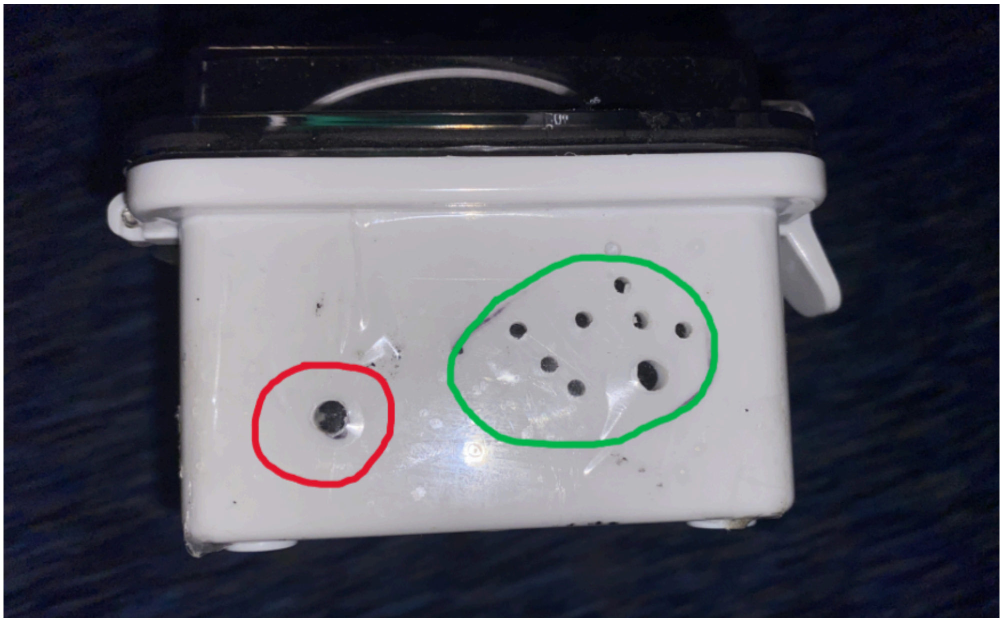


WATERPROOF ENCLOSURE MODIFICATIONS

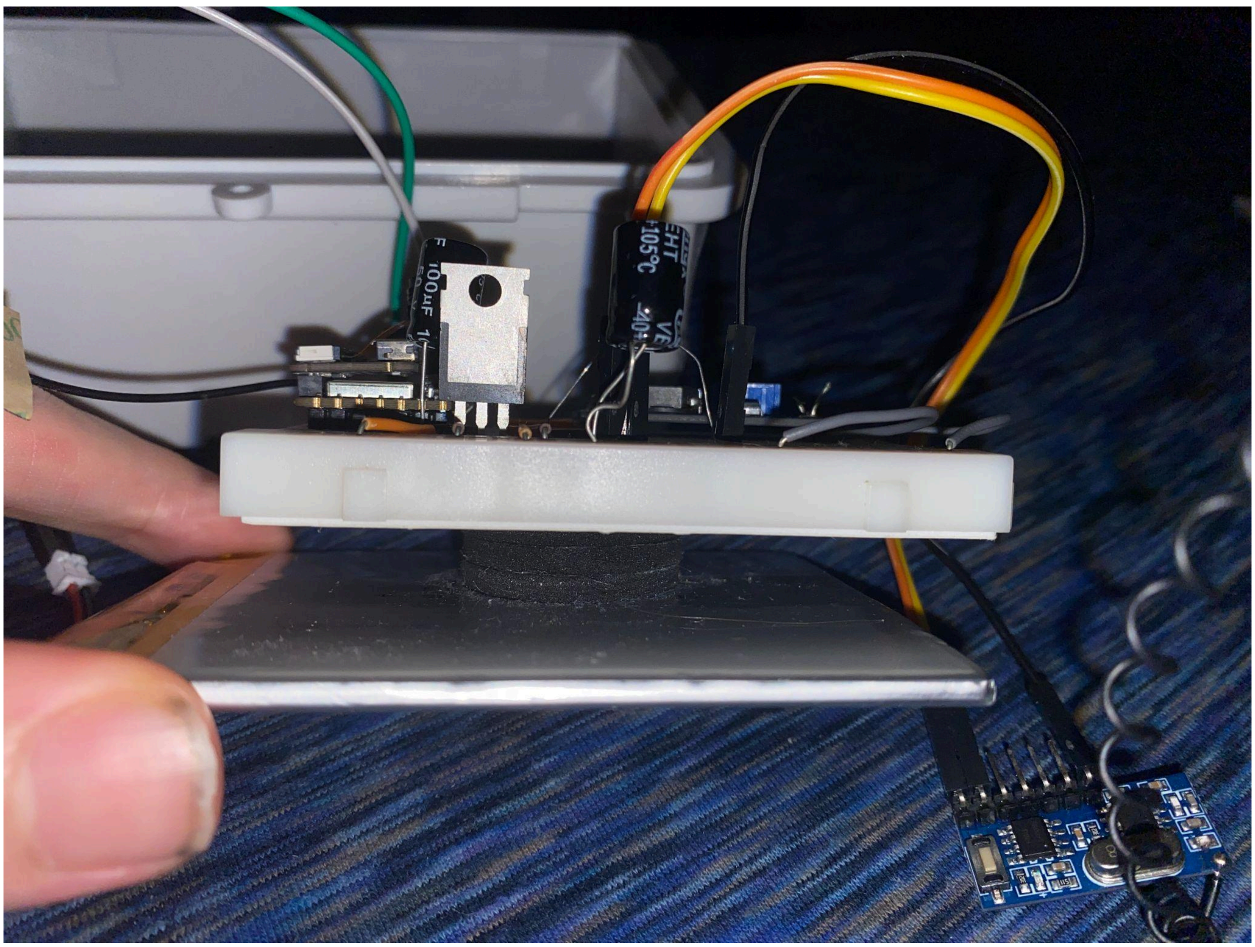
Red Circle = A hole made to fit solar panel cable through.

Green Circle = Vent holes were supposed to be covered with high thermal conductivity water resistant material but after testing, the max internal temperature in the enclosure, with full power, out in the highest temperature of a summer day, was well below harmful levels so I just taped it off for max moisture resistance.

Tape is also covering the solar panel hole because the camera ended up being so efficient I didn't need to charge it for a month at a time.



PRIMARY MODULE ATTACHED TO BATTERY WITH STICKY PADS



HOW IT WORKS

1. **The ESP32S3 is coded in Arduino IDE. The primary file is a .ino file. The code is on my GitHub (https://github.com/tddev123/Camron/tree/main). The code sets up the pin configurations, the WiFi functionality, the basic camera initialization settings, and functions. You can add custom loop functions to make the camera do timed tasks automatically. I left some of the extra code there as an example for anyone who might want to use it. For the purpose of this project. The code is set up for continuous recording.**

2. **The ESP32S3 is powered through its 5V pin. The 3.7V 3000mAh LiPo battery is connected to the MT3608 boost converter to output 5V. The output of the boost converter is connected to the power rails of the breadboard. A wire connects the + power rail directly to the ESP32S3 5V pin. The ground pin of the ESP32S3 is connected to the drain of the RFP30N06LE Mosfet. The source of the Mosfet is connected to the - power rail. The Gate of the Mosfet is connected to the Data output pin of the RX480E receiver module.**

3. **The TX118SA-4 transmitter module is paired with the RX480E receiver module. On a separate breadboard, the TX118SA-4 transmitter is powered by anything that can safely supply 3.3-5V. When the TX118SA-4 is powered and one of the K pins (K3) is connected to ground, the module begins transmitting a 433 MHz RF signal. This signal is then picked up by the RX480E receiver module, causing it to output a logic high signal out of its corresponding Data pin (D3). The output pin of the RX480E being used is the VT pin which will output high when any of the TX118SA-4's K pins are connected to ground, making it simpler to use.**

4. **When the RX480E Data pin (VT) outputs high, it applies a voltage to the Mosfets gate that is equal to the voltage powering the RX480E. This then completes the connection of ESP32S3's ground pin to the - power rail. The ESP32S3 then turns on and begins its video stream which can be viewed through its own self hosted local HTTP web server which can be viewed by going to the URL specified in the code file.**

5. **Disconnecting the TX118SA-4's K pin will disable the transmission and remove the voltage on the Mosfet gate, cutting the power to the ESP32S3, turning it off.**

Questions?

Why is the RF Receiver Antenna Stretched out?

Because, through testing, I found that it has more range like that.

Why is a resistor used to connect the RX480 output pin to the Mosfet?

Because it reduces current consumption and still works.

What's the point of this project?

This was my first ever electrical engineering project. I started this project in my first month of electrical engineering because I believe the fastest way to learn something is to dive in and make mistakes.

What was the biggest problem you solved when building this?

There were a lot of big problems but what I would consider to be the cleverest solution was using the Mosfet and the RF module to remotely activate and deactivate the camera. I did not learn about that from anywhere, it was a purely creative solution I came up with to solve the problem of the ESP32S3 using 500ma nominally non-stop if the camera and web server are left on. This allows me to be able to access the camera anytime I want and extends the battery life tremendously.

The ESP32S3 uses about 500mA nominally and can peak as high as 700mA. This created 2 main issues. My initial design used a 1,000mAh battery and a TP4056 charging module. When the ESP32S3 would pull peak currents, it would surpass what the battery (running through the TP4056) could supply and cause brown outs/black outs. Replacing the battery with a 3000mAh battery and replacing the TP4056 with a BQ25606 solved this problem completely.